

Claude Multi-Account Fine Tuning

Running multiple Claude Code accounts on one host with shared history, memory, and settings

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Claude Multi-Account Fine Tuning

This document captures, in reproducible detail, how a single host can run multiple Claude Code accounts (e.g. `claude1`, `claude2`, ..., `claudeN`) while sharing conversation history, project memory, todos, plans, plugins, and settings between them. Only account identity (`.credentials.json`, `.claude.json`, MCP auth cache) is kept private per account.

Companion artifacts:

- `~/Documents/scripts/` — automation scripts (the source of truth for the procedure described here).
- `Claude_Multi_Account_Fine_Tuning.html` — same content as this file, rendered as a standalone HTML page.
- `Claude_Multi_Account_Fine_Tuning.pdf` — same content, paginated PDF.

The HTML and PDF are regenerated by `claude-export-docs.sh`, so editing the markdown is the only place changes need to be made.

1. Motivation

The Claude Code CLI is invoked as `claude`. By default it writes all of its state — credentials, conversation transcripts, project memory, todos, plans, plugin installs, settings — under a single config directory (`~/.claude` on Linux/macOS).

When you want to use **multiple Anthropic accounts** (for example, one personal Max subscription and one team/work account), you typically run two separate config dirs and switch between them with an environment variable:

```
# ~/.bashrc (before this setup)
export CLAUDE_BIN="$HOME/.local/bin/claude"
alias claude1="CLAUDE_CONFIG_DIR=$HOME/.claude-milos85vasic
               $CLAUDE_BIN"
alias claude2="CLAUDE_CONFIG_DIR=$HOME/.claude-milos85vasic2nd
               $CLAUDE_BIN"
```

That works for isolation, but the two accounts now have **completely separate** project memory, history, todos, and plugin sets. Switching accounts means losing the working context you built up under the other.

Goal: keep the two-alias UX (cheap account switch) while ensuring both accounts read and write the same project memory, conversation transcripts, todos, plans, plugins, and settings.

2. How Claude Code stores state

Every key piece of state lives under the directory pointed at by CLAUDE_CONFIG_DIR (with ~/.claude as the default). Items fall into three categories:

Category	Path under CLAUDE_CONFIG_DIR	What it is
Identity (private)	.credentials.json	OAuth tokens for the Anthropic account.
	.claude.json	Account onboarding/migration markers, numStartups, etc.
	mcp-needs-auth-cache.json	Per-account MCP auth state.
Work product (share)	projects/<slug>/<uuid>.jsonl	Per-session conversation transcript.
	projects/<slug>/memory/*.md	Project memory entries (what Skill auto-memory writes).
	history.jsonl	Prompt-line history (Up/Down recall).
	todos/, tasks/, plans/	UUID-named files for TaskCreate/Plan tooling.
	file-history/	Snapshots of files edited via the Edit tool.
	paste-cache/	Long-paste buffer storage.
	plugins/	

Category	Path under CLAUDE_CONFIG_DIR	What it is
Runtime (share-able with care)		Installed plugins, marketplaces, manifest JSON.
	settings.json	Top-level config: enabledPlugins, effortLevel, etc.
	CLAUDE.md	User-scope memory / instructions.
	sessions/	Active session lockfiles. Not safe to share between concurrent processes.
	telemetry/	Per-account telemetry batches.
	cache/, backups/, session-env/, shell- snapshots/	Ephemeral runtime caches.
	stats-cache.json	Per-account usage / quota stats.

The key observation: every **work-product** item is keyed either by UUID (session, todo, plan, file edit) or by project slug. That means combining two accounts' work-product directories is a **set-union operation**: copying both into the same destination never overwrites either's data, because each file's path is already unique.

The handful of files that genuinely need merging are:

- settings.json — both files have a top-level enabledPlugins map. We union it, then keep the most permissive top-level keys (effortLevel: "high", skipDangerousModePermissionPrompt: true).
- history.jsonl — append-only prompt log. We concatenate both files and de-duplicate by line, preserving order of first appearance.
- plugins/installed_plugins.json and plugins/known_marketplaces.json — these embed **absolute paths** to plugin code. We rewrite those paths to live under the shared store regardless of which account they came from.

3. Architecture after fine-tuning

```
~/.claude-shared/          <-- single source of truth
  projects/                (transcripts + memory; per-slug)
  todos/  tasks/  plans/
```

```

file-history/  paste-cache/
plugins/
  installed_plugins.json      (paths rewritten -> ~/.claude-
shared/...)
  known_marketplaces.json    (paths rewritten -> ~/.claude-
shared/...)
  cache/<marketplace>/<plugin>/<version>/
  marketplaces/<marketplace>/
shell-snapshots/  session-env/
telemetry/  sessions/  backups/  cache/
history.jsonl  settings.json  stats-cache.json
CLAUDE.md

```

```

~/.claude-milos85vasic/      <-- account 1 (claude1 alias)
  .credentials.json          (PRIVATE – claude1's OAuth tokens)
  .claude.json               (PRIVATE – claude1's onboarding state)
  mcp-needs-auth-cache.json  (PRIVATE)
  projects                   -> ~/.claude-shared/projects
  todos                       -> ~/.claude-shared/todos
  tasks                       -> ~/.claude-shared/tasks
  plans                       -> ~/.claude-shared/plans
  plugins                     -> ~/.claude-shared/plugins
  settings.json              -> ~/.claude-shared/settings.json
  history.jsonl              -> ~/.claude-shared/history.jsonl
  CLAUDE.md                  -> ~/.claude-shared/CLAUDE.md
  ... (every shared item is a symlink)

```

```

~/.claude-milos85vasic2nd/   <-- account 2 (claude2 alias)
  .credentials.json          (PRIVATE – claude2's OAuth tokens)
  .claude.json               (PRIVATE – claude2's onboarding state)
  mcp-needs-auth-cache.json  (PRIVATE)
  ... (same symlink set as above)

```

```

~/.claude/                   <-- user-scope default root
  CLAUDE.md                  -> ~/.claude-shared/CLAUDE.md
  plugins/
    cache                     -> ~/.claude-shared/plugins/cache
    marketplaces              -> ~/.claude-shared/plugins/marketplaces

```

Properties of this layout:

- 1. Account switching is purely a credentials swap.** Running `claude1` vs `claude2` changes `CLAUDE_CONFIG_DIR`, which changes which `.credentials.json` is loaded — everything else (history, memory, plugins) resolves through symlinks to the same shared store.
- 2. Adding a new account is the same operation, parameterised.** A new `~/.claude-<name>` dir with the same symlink set + its own `.credentials.json` slots in seamlessly. See `claude-add-account.sh`.

3. **Account identity stays compartmentalised.** OAuth tokens are not in the shared store, so leaking the shared store does not leak account credentials.
 4. **Rollback is mechanical.** Every original directory was moved to `<path>.preunify.<timestamp>` before the symlink replaced it. A single script (`claude-rollback.sh`) reverses the migration.
-

4. Reproducing this on any host

This section is the actual step-by-step procedure. It assumes you have two (or more) Claude Code accounts already set up the conventional way — one config directory per account, each with its own `.credentials.json`. If you don't have a second account yet, run the first three steps and then use `claude-add-account.sh` to create one.

4.1 Prerequisites

```
# Linux (Debian/Ubuntu/Fedora/Arch)
sudo apt-get install -y rsync jq gawk pandoc weasyprint      #
    Debian/Ubuntu

# or
sudo dnf install -y rsync jq gawk pandoc weasyprint          #
    Fedora

# or
sudo pacman -S rsync jq gawk pandoc weasyprint              # Arch

# macOS (Homebrew)
brew install rsync jq gawk pandoc weasyprint
```

`weasyprint` is optional but produces the best-looking PDF; the export script will fall back to `wkhtmltopdf` or `chromium --headless` if `weasyprint` isn't available.

4.2 Lay down the scripts directory

```
mkdir -p ~/Documents/scripts
# Copy or git-clone the contents of this repo's scripts/ into ~/Documents/scripts.

# At minimum you need:
#   lib.sh
#   install.sh
#   claude-unify.sh
#   claude-add-account.sh
#   claude-remove-account.sh
#   claude-list-accounts.sh
#   claude-rollback.sh
```

```
# claude-export-docs.sh
chmod +x ~/Documents/scripts/*.sh
```

4.3 Run the installer

```
bash ~/Documents/scripts/install.sh
```

What `install.sh` does, in order:

1. Verifies `rsync`, `jq`, `awk` are on `PATH`; warns if `pandoc` is missing.
2. Symlinks every `claude-*.sh` script into `~/.local/bin/` so they're invokable as plain commands (`claude-unify`, `claude-add-account`, ...).
3. Adds `export PATH="$HOME/.local/bin:$PATH"` to `~/.bashrc` and `~/.zshrc` if missing.
4. Creates the managed alias file at `~/.local/share/claude-multi-account/aliases.sh` and ensures it's sourced from `~/.bashrc` and `~/.zshrc`.
5. Migrates any pre-existing inline alias `claude<N>=...` lines from `~/.bashrc` / `~/.zshrc` into the managed alias file, commenting out the originals (and saving a `.preunify.<timestamp>` backup of the rc file first).
6. Auto-detects every `~/.claude-*` directory and runs `claude-unify.sh` to merge them into `~/.claude-shared/`.
7. Re-renders the PDF and HTML from this markdown via `claude-export-docs.sh`.

After it finishes, reload your shell:

```
exec $SHELL -l
claude-list-accounts
```

You should see every account dir, with `CREDs: yes` and a `LINKs: 6/6` (or higher) indicator showing the shared symlinks are in place.

4.4 Add a third (or fourth, ...) account

```
claude-add-account
# prompts:
# Alias name [claude3]:
# Config directory [/home/you/.claude-claude3]:
```

If you accept both defaults, you get:

- `~/.claude-claude3/` created with the full set of shared-state symlinks already in place.
- `alias claude3="CLAUDE_CONFIG_DIR=$HOME/.claude-claude3 \ $CLAUDE_BIN"` appended to the managed alias file.
- A reminder to authenticate: `claude3 /login`.

Custom alias names work just as well:

```
claude-add-account --alias work --dir ~/.claude-work --yes
work /login
```

4.5 Verify the result

```
claude-list-accounts
```

```
# Expected output (your alias names may differ):
# ALIAS      CONFIG DIR
#          CRED$  LINK$  NOT$
# claude1    /home/you/.claude-milos85vasic
#          yes    6/6
# claude2    /home/you/.claude-milos85vasic2nd
#          yes    6/6
# claude3    /home/you/.claude-claude3
#          no     6/6
#
# Shared store: /home/you/.claude-shared
# Alias file:   /home/you/.local/share/claude-multi-account/
#               aliases.sh
```

You can confirm symlinks resolve correctly:

```
readlink -f ~/.claude-milos85vasic/projects
# /home/you/.claude-shared/projects

readlink -f ~/.claude-milos85vasic2nd/projects
# /home/you/.claude-shared/projects

# Both should print the same path. The same is true for todos,
# plans,
# settings.json, history.jsonl, CLAUDE.md, etc.
```

4.6 Rollback if needed

Every directory or file the unifier touched was moved to `<path>.preunify.<timestamp>` before the symlink was created. To restore:

```
claude-rollback
# moves ~/.claude-shared aside and restores every .preunify.*
# backup
```

The shared store isn't deleted — it's renamed to `~/.claude-shared.removed.<timestamp>` so you can manually recover any data written through the symlinks before rolling back.

5. Operational notes

5.1 Concurrent use of two accounts

sessions/ is shared in this configuration. A live Claude Code session holds its lockfile under sessions/, and two concurrent processes writing to the same lockfile path can confuse each other. **Avoid running claude1 and claude2 simultaneously in two terminals.** If you need true concurrent use, override the unifier to keep sessions/ private:

```
# in claude-unify.sh, remove "sessions" from SHARED_ITEMS
# then run claude-unify again – it'll relink sessions/ as per-account
```

history.jsonl is append-only and is mostly safe under concurrent writes (worst case: a small number of duplicate lines, which the next unify run will de-dupe).

5.2 Plugin path rewriting

Claude Code records absolute paths in plugins/installed_plugins.json and plugins/known_marketplaces.json. These paths can point to one of:

- ~/.claude/plugins/cache/... (user-scope installs from any account)
- ~/.claude-<name>/plugins/cache/... (project-scope installs from that account)

The unifier rsyncs plugin code from all three roots (~/.claude/plugins/, plus every account's plugins/) into ~/.claude-shared/plugins/, then rewrites every installPath / installLocation to point under ~/.claude-shared/plugins/. It also symlinks ~/.claude/plugins/cache and ~/.claude/plugins/marketplaces into the shared store, so any future user-scope install lands in the unified spot.

If you ever install a plugin and Claude reports "plugin code not found," re-run claude-unify.sh to repath the manifest.

5.3 Adding a new account from scratch (no existing dir)

The claude-add-account.sh script works whether or not the target dir exists yet. If it doesn't exist, the script creates an empty dir, links every shared item into it, and registers the alias. The new account has no credentials of its own — your next step is claudeN / login.

If you want to also pre-populate the new account with a `.claude.json` template (e.g., to skip the new-user warmup tips), copy one from an existing account *after* the add-account script finishes:

```
cp ~/.claude-milos85vasic2nd/.claude.json ~/.claude-
  claude3/.claude.json
# but DO NOT copy .credentials.json – that would leak the source
# account's OAuth tokens to the new account
```

5.4 Removing an account

```
claude-remove-account --alias claude3
# default: archives ~/.claude-claude3 -> ~/.claude-
  claude3.removed.<timestamp>
# add --delete to remove it instead
```

The shared store is left intact; other accounts continue to function.

5.5 Backups created during install

The installer never deletes existing data. Every directory or file it replaced lives at `<original-path>.preunify.<timestamp>` (or `<original-path>.removed.<timestamp>` for rollback). You can safely delete those backups once you've confirmed the new setup works:

```
find ~ -maxdepth 3 -name '*.preunify.*' -print      # see what's
  there
find ~ -maxdepth 3 -name '*.preunify.*' -delete     # delete once
  happy
```

A pre-migration tarball is also placed in `~/Documents/claude-config-backups/` by the original migration. Keep it until you're confident in the new layout.

6. Cross-platform reproduction

Linux

Everything in this document was developed and tested on Linux 6.12 with bash 5.2, jq 1.8.1, rsync 3.2.7, pandoc 3.9. No Linux-specific behaviours are relied on; the scripts pass shellcheck clean and use only POSIX-portable tools plus jq.

macOS

macOS ships an older bash (3.2) by default. Install GNU bash via Homebrew (`brew install bash`) and update the shebangs only if you hit syntax issues — the scripts use bash 4+ features only where the fallback is explicit (mapfile, associative arrays). On a default macOS:

```
brew install bash jq rsync gawk pandoc weasyprint
echo "$(brew --prefix)/bin/bash" | sudo tee -a /etc/shells
chsh -s "$(brew --prefix)/bin/bash"
```

Everything else (paths, \$HOME, ~/.claude-* convention) is identical to Linux.

WSL / Windows Subsystem for Linux

WSL is just Linux for our purposes. Use the Linux instructions. If you also run Claude Code natively on Windows, the Windows version uses a different config root entirely (%APPDATA%\claude) and is not covered by this toolkit.

7. Tests

Everything in this toolkit is covered by a hermetic test suite under ~/Documents/scripts/tests/. Each test runs against a temporary \$HOME created by `mktemp -d -t cma-test.XXXX`; the real ~/.claude-* directories are never touched. After the test process exits, the sandbox is `rm -rf'd` (with a safety check that the path basename starts with `cma-test.`).

7.1 Running the suite

```
# All tests:
bash ~/Documents/scripts/tests/run-all.sh

# A single file (no test_prefix, no .sh suffix):
bash ~/Documents/scripts/tests/run-all.sh unify
bash ~/Documents/scripts/tests/run-all.sh add_remove
```

Output is line-based:

```
==> test_unify.sh
```

- every shared item becomes a symlink into the shared store
[PASS] .claude-acct1 projects linked: ...
[PASS] .claude-acct2 projects linked: ...
- settings.json enabledPlugins is the union {a,b,c}
[PASS] union (.enabledPlugins | keys | sort | join(",") =

a,b,c)

...

✓ 33 passed, 0 failed

The orchestrator's exit code is 0 only if every test_*.sh file's own summary returned 0.

7.2 What each test covers

File	What it asserts
test_lib.sh	cma_validate_alias accepts/rejects correctly; cma_suggest_alias returns the next free claudeN (skipping past existing aliases, not just counting); cma_write_alias is idempotent; cma_remove_alias deletes; cma_ensure_alias_file patches .bashrc; cma_detect_accounts excludes the shared store.
test_unify.sh	Every shared item becomes a symlink to \$SHARED_DIR; settings.json enabledPlugins is a strict union of every account's set; non-enabledPlugins top-level keys are preserved with right-most file winning; history.jsonl is concatenated and de-duplicated; on memory-file conflicts, the most recently touched account wins; plugins/installed_plugins.json and known_marketplaces.json have absolute paths rewritten to \$SHARED_DIR/plugins/; .credentials.json and .claude.json are NOT symlinked; re-running claude-unify.sh produces a byte-identical settings.json (idempotency); N>2 accounts (adding acct3 and re-unifying) works; --rollback restores every .preunify.* backup and moves the shared store aside.
test_add_remove.sh	claude-add-account.sh --yes creates the dir with full symlink set and writes a shell-safe alias line; refuses to overwrite an existing dir; accepts a custom alias name + custom dir; validates alias names against the safe regex; claude-remove-account.sh --archive moves the dir aside and removes the alias; --delete actually deletes; unknown alias errors out.
test_list.sh	Empty host (no accounts) renders just the header; populated host renders one row per detected account dir; the shared store path

File	What it asserts
	is printed; accounts missing credentials show CREDs: no.

7.3 How the test harness works

Two small libraries sit under tests/lib/:

- **assert.sh** — `it`, `assert_eq`, `assert_file`, `assert_dir`, `assert_symlink_to`, `assert_not_symlink`, `assert_file_contains`, `assert_file_not_contains`, `assert_jq`, `assert_lines`, `assert_exit`. Each helper prints a [PASS] or [FAIL] line and adjusts the counters used by summary at end-of-file.
- **sandbox.sh** — `make_sandbox` (creates a tempdir and overrides `HOME`, `SHARED_DIR`, `ALIAS_FILE`, `DEFAULT_DIR`, `CLAUDE_BIN`), `make_account NAME [--plugins] [--settings JSON] [--history a|b|c] [--memory KEY:VALUE] [--todo SUBJECT]` (builds a realistic per-account fixture dir), plus thin wrappers `run_unify`, `run_add_account`, etc.

One subtlety worth knowing: `lib.sh` (the production helper) has `set -euo pipefail`. When a test sources it for unit-testing the helper functions, the `set -e` flag stays active in the test shell. The unit test file (`test_lib.sh`) therefore runs `set +e` immediately after sourcing `lib.sh` so that intentional failures (e.g. `cma_validate_alias "bad name"`) don't abort the whole test process. End-to-end tests don't have this problem because they invoke the scripts as external processes (via `run_unify`, etc.), which run in their own shell.

7.4 Adding a new test

1. Create `tests/test_<feature>.sh`. Mark it executable.
2. Source the two libs:

```
TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"
source "$TESTS_DIR/lib/assert.sh"
source "$TESTS_DIR/lib/sandbox.sh"
```

3. Call `make_sandbox` once at the top.
4. Use it `"..."` to label each scenario, then a sequence of `assert_*` calls.
5. End the file with `summary` so the orchestrator picks up the result.
6. Re-run `bash tests/run-all.sh` — auto-discovery will pick it up.

8. The complete script source

For convenience and audit, the full source of the toolkit is reproduced below. The authoritative copies live at ~/Documents/scripts/.

8.1 lib.sh

```
#!/usr/bin/env bash
# lib.sh – shared functions for the Claude multi-account toolkit.
# Sourced by the other scripts; no side effects on its own.

set -euo pipefail

# Resolve paths the user can override. SHARED_DIR is the single
# canonical
# location for cross-account state; ALIAS_FILE is the rc-sourced
# file we
# manage aliases through; ACCOUNT_PREFIX is the dir-name prefix
# for new
# per-account config directories.
: "${SHARED_DIR:= $HOME/.claude-shared}"
: "${ALIAS_FILE:= $HOME/.local/share/claude-multi-account/
aliases.sh}"
: "${ACCOUNT_PREFIX:= .claude-}"

CLAUDE_BIN_DEFAULT="${CLAUDE_BIN:- $HOME/.local/bin/claude}"

cma_log() { printf '\033[36m[ cma ]\033[0m %s\n' "$*" >&2; }
cma_warn() { printf '\033[33m[ cma warn ]\033[0m %s\n' "$*" >&2; }
cma_err() { printf '\033[31m[ cma err ]\033[0m %s\n' "$*" >&2; }
cma_die() { cma_err "$*"; exit 1; }

cma_require() {
  command -v "$1" >/dev/null 2>&1 || cma_die "missing required
  tool: $1"
}

# Detect Linux vs macOS for platform-specific commands.
cma_os() {
  case "$(uname -s)" in
    Linux*) echo linux ;;
    Darwin*) echo macos ;;
    *) echo unknown ;;
  esac
}

# Find all existing Claude account config directories under $HOME,
# matching
# the convention `.claude-<name>`. Echoes absolute paths, one per
# line,
```

```

# sorted. The default `~/.claude` is intentionally excluded
# because we treat
# it as the shared user-scope spot, not an account dir.
cma_detect_accounts() {
    find "$HOME" -maxdepth 1 -type d -name "${ACCOUNT_PREFIX}*" 2>/
        dev/null \
        | grep -v -- "-shared\$" \
        | sort
}

# Read all aliases of the form `alias name=...` from $ALIAS_FILE
# and print
# their names (one per line). Returns nothing if the file is
# absent.
cma_existing_aliases() {
    [[ -f "$ALIAS_FILE" ]] || return 0
    awk -F'[ =]+' '/^[[:space:]]*alias[[:space:]]+/{print $2}'
        "$ALIAS_FILE"
}

# Suggest the next free `claude<N>` alias by scanning current
# aliases.
# E.g., if claude1 and claude2 exist, returns "claude3".
cma_suggest_alias() {
    local highest=0 n
    for a in $(cma_existing_aliases); do
        if [[ "$a" =~ ^claude([0-9]+)$ ]]; then
            n="${BASH_REMATCH[1]}"
            (( n > highest )) && highest="$n"
        fi
    done
    echo "claude$((highest + 1))"
}

# Validate that an alias name is safe to embed in a generated
# `alias ...=`
# line – strictly alphanumeric + underscore/hyphen, starting with
# a letter.
cma_validate_alias() {
    [[ "$1" =~ ^[a-zA-Z][a-zA-Z0-9_-]*$ ]] || cma_die
        "invalid alias name: $1"
}

# Ensure $ALIAS_FILE exists with the header sentinel and is
# sourced from
# the user's rc files. Idempotent – safe to call repeatedly.
cma_ensure_alias_file() {
    mkdir -p "$(dirname "$ALIAS_FILE")"
    if [[ ! -f "$ALIAS_FILE" ]]; then
        cat > "$ALIAS_FILE" <<EOF
# Managed by claude-multi-account. Do not edit by hand; use
# ~/Documents/scripts/claude-add-account.sh to add accounts.
export CLAUDE_BIN="${CLAUDE_BIN_DEFAULT}"
    fi
}

```

```

EOF
fi
local rc src_line="source \"$ALIAS_FILE\"
for rc in "$HOME/.bashrc" "$HOME/.zshrc"; do
    [[ -f "$rc" ]] || continue
    if ! grep -F -q "$src_line" "$rc"; then
        printf '\n# Claude multi-account aliases\n%s\n' "$src_line"
        >> "$rc"
        cma_log "added source line to $rc"
    fi
done
}

# Add (or refresh) a single alias entry in $ALIAS_FILE.
# Idempotent.
cma_write_alias() {
    local alias_name="$1" config_dir="$2"
    cma_validate_alias "$alias_name"
    cma_ensure_alias_file
    # Strip any prior line for this alias, then append the new one.
    local tmp; tmp="$(mktemp)"
    grep -v -E "^alias[[:space:]]+${alias_name}=" "$ALIAS_FILE" >
        "$tmp" || true
    printf 'alias %s="CLAUDE_CONFIG_DIR=%s \\$CLAUDE_BIN"\n' \
        "$alias_name" "$config_dir" >> "$tmp"
    mv "$tmp" "$ALIAS_FILE"
}

# Remove an alias line. Idempotent.
cma_remove_alias() {
    local alias_name="$1"
    [[ -f "$ALIAS_FILE" ]] || return 0
    local tmp; tmp="$(mktemp)"
    grep -v -E "^alias[[:space:]]+${alias_name}=" "$ALIAS_FILE" >
        "$tmp" || true
    mv "$tmp" "$ALIAS_FILE"
}

```

8.2 claude-unify.sh

```

#!/usr/bin/env bash
# claude-unify.sh – Unify N Claude Code account config dirs into a
#   single
# shared store so every account sees the same projects/
#   conversations,
# memory, history, todos, plans, plugins, and settings.
#
# Inputs:
#   * Positional args: one or more account config directories. If
#     none
#     are given, all `~/.claude-*` directories are auto-detected
#     (excluding

```



```
#     the shared store itself).
#     * Env vars:
#         SHARED_DIR      target shared store (default: ~/.claude-
#                           shared)
#         DEFAULT_DIR     user-scope plugin root (default: ~/.claude)
#
# Use --rollback to restore the .preunify.* backups and remove the
# shared
# store. Safe to re-run any time.
```

```
set -euo pipefail
```

```
LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
```

```
# shellcheck source=lib.sh
```

```
source "$LIB_DIR/lib.sh"
```

```
DEFAULT_DIR="${DEFAULT_DIR:-$HOME/.claude}"
```

```
cma_require rsync
```

```
cma_require jq
```

```
cma_require awk
```

```
# Items each account dir contributes to the shared store. Order
# matters
```

```
# only insofar as we want plugins last so the path-rewrite step
# has all
```

```
# manifest entries to rewrite at once.
```

```
SHARED_ITEMS=(
```

```
    projects
```

```
    todos
```

```
    tasks
```

```
    plans
```

```
    file-history
```

```
    paste-cache
```

```
    shell-snapshots
```

```
    session-env
```

```
    telemetry
```

```
    sessions
```

```
    backups
```

```
    cache
```

```
    stats-cache.json
```

```
    history.jsonl
```

```
    settings.json
```

```
    plugins
```

```
)
```

```
PRIVATE_ITEMS=(
```

```
    .credentials.json
```

```
    .claude.json
```

```
    mcp-needs-auth-cache.json
```

```
)
```

```

ts() { date +%Y%m%d%H%M%S; }

already_linked_to_shared() {
    [[ -L "$1" ]] || return 1
    [[ "$(readlink -f "$1")" == "$(readlink -f "$2" 2>/dev/null ||
        echo "$2")" ]]
}

backup_and_remove() {
    local p="$1"
    [[ -e "$p" || -L "$p" ]] || return 0
    mv "$p" "${p}.preunify.${ts}"
}

link_to_shared() {
    local item="$1" acct
    for acct in "${ACCOUNTS[@]}; do
        [[ -d "$acct" ]] || continue
        local target="$acct/$item"
        already_linked_to_shared "$target" "$SHARED_DIR/$item" &&
            continue
        backup_and_remove "$target"
        mkdir -p "$(dirname "$target")"
        ln -s "$SHARED_DIR/$item" "$target"
    done
}

merge_dir_into_shared() {
    local item="$1" acct
    mkdir -p "$SHARED_DIR/$item"
    # First pass: each account fills only gaps (--ignore-existing)
    # so the
    # union is preserved across N accounts.
    for acct in "${ACCOUNTS[@]}; do
        if [[ -d "$acct/$item" && ! -L "$acct/$item" ]]; then
            rsync -a --ignore-existing "$acct/$item/" "$SHARED_DIR/
                $item/"
        fi
    done
    # Second pass: the LAST account (assumed most recently active)
    # overlays
    # its versions for files that exist in multiple accounts. This
    # biases
    # towards the freshest source for conflicting files like memory/
    # *.md.
    local last="${ACCOUNTS[-1]}"
    if [[ -d "$last/$item" && ! -L "$last/$item" ]]; then
        rsync -a "$last/$item/" "$SHARED_DIR/$item/"
    fi
}

merge_history_jsonl() {
    local acct tmp; tmp="$(mktemp)"

```

```

for acct in "${ACCOUNTS[@]}"; do
    local f="$acct/history.jsonl"
    if [[ -f "$f" && ! -L "$f" ]]; then cat "$f" >> "$tmp"; fi
done
if [[ -f "$SHARED_DIR/history.jsonl" && ! -L "$SHARED_DIR/
history.jsonl" ]]; then
    cat "$SHARED_DIR/history.jsonl" >> "$tmp"
fi
if [[ -s "$tmp" ]]; then
    awk 'NF && !seen[$0]++' "$tmp" > "$SHARED_DIR/history.jsonl"
else
    : > "$SHARED_DIR/history.jsonl"
fi
rm -f "$tmp"
return 0
}

merge_settings_json() {
    local acct files=() resolved
    for acct in "${ACCOUNTS[@]}"; do
        local f="$acct/settings.json"
        if [[ -L "$f" ]]; then
            resolved="$(readlink -f "$f")"
            if [[ -f "$resolved" ]]; then files+=("$resolved"); fi
        elif [[ -f "$f" ]]; then
            files+=("$f")
        fi
    done
    # De-duplicate. On re-runs both account symlinks resolve to the
    # same
    # shared file, and passing the same path twice plus redirecting
    # back to
    # it truncates the file before jq reads it.
    if (( ${#files[@]} > 1 )); then
        mapfile -t files << (printf '%s\n' "${files[@]}" | awk '!
seen[$0]++')
    fi
    case "${#files[@]}" in
        0) return 0 ;;
        1) cp -p "${files[0]}" "$SHARED_DIR/settings.json.tmp"
            mv "$SHARED_DIR/settings.json.tmp" "$SHARED_DIR/
settings.json" ;;
        *)
            # Right-most file's top-level keys win, except
            # enabledPlugins which
            # is the union across all accounts (any "true" survives).
            # We bind the slurped array to $all so the inner reductions
            # don't
            # iterate over the values of the partially-merged outer
            # object.
            # Write to a temp file first so we never truncate-then-read
            # shared.
            jq -s '

```

```

        . as $all
        | (reduce $all[] as $x ({}; . * $x))
        | .enabledPlugins = (reduce $all[] as $x ({}; . +
            ($x.enabledPlugins // {})))
        ' "${files[@]}" > "$SHARED_DIR/settings.json.tmp"
mv "$SHARED_DIR/settings.json.tmp" "$SHARED_DIR/
settings.json"

    ;;
esac
return 0
}

merge_file_into_shared() {
    local item="$1" acct newest=""
    for acct in "${ACCOUNTS[@]}; do
        local f="$acct/$item"
        if [[ -L "$f" ]]; then continue; fi
        if [[ ! -f "$f" ]]; then continue; fi
        if [[ -z "$newest" || "$f" -nt "$newest" ]]; then
            newest="$f"; fi
    done
    if [[ -n "$newest" ]]; then cp -p "$newest" "$SHARED_DIR/
$item"; fi
    return 0
}

absorb_default_plugins() {
    [[ -d "$DEFAULT_DIR/plugins" && ! -L "$DEFAULT_DIR/plugins" ]]
    || return 0
    mkdir -p "$SHARED_DIR/plugins"
    rsync -a --ignore-existing "$DEFAULT_DIR/plugins/" "$SHARED_DIR/
plugins/"
}

rewrite_plugin_paths() {
    local f="$SHARED_DIR/plugins/installed_plugins.json"
    if [[ -f "$f" ]]; then
        local tmp; tmp="$(mktemp)"
        jq --arg shared "$SHARED_DIR" '
            .plugins |= with_entries(
                .value |= map(
                    if ((.installPath // "") | test(".*plugins/cache/"))
                    then .installPath = ($shared + "/plugins/cache/" +
                        (.installPath | sub(".*plugins/cache/"; "")))
                    else . end
                )
            )' "$f" > "$tmp" && mv "$tmp" "$f"
    fi
    local km="$SHARED_DIR/plugins/known_marketplaces.json"
    if [[ -f "$km" ]]; then
        local tmp; tmp="$(mktemp)"
        jq --arg shared "$SHARED_DIR" '
            with_entries(

```

```

        if ((.value.installLocation // "") | test(".*plugins/
        marketplaces/"))
        then .value.installLocation = ($shared + "/plugins/
        marketplaces/" + (.value.installLocation | sub(".*
        plugins/marketplaces/"; "")))
        else . end
    )' "$km" > "$tmp" && mv "$tmp" "$km"
fi
}

# Repoint ~/.claude/plugins/cache and ~/.claude/plugins/
# marketplaces at
# the shared store so future user-scope installs land in one
# place.
link_default_plugin_subdirs() {
    for sub in cache marketplaces; do
        local src="$SHARED_DIR/plugins/$sub" tgt="$DEFAULT_DIR/
        plugins/$sub"
        [[ -d "$src" ]] || continue
        already_linked_to_shared "$tgt" "$src" && continue
        mkdir -p "$DEFAULT_DIR/plugins"
        backup_and_remove "$tgt"
        ln -s "$src" "$tgt"
    done
}

# CLAUDE.md (user-scope memory) lives at ~/.claude/CLAUDE.md by
# default.
# We promote it to the shared store and symlink all three
# locations (each
# account dir + ~/.claude) so any account writing user memory
# updates the
# same file.
sync_claude_md() {
    local shared_md="$SHARED_DIR/CLAUDE.md" src
    if [[ ! -f "$shared_md" ]]; then
        if [[ -f "$DEFAULT_DIR/CLAUDE.md" && ! -L "$DEFAULT_DIR/
        CLAUDE.md" ]]; then
            cp -p "$DEFAULT_DIR/CLAUDE.md" "$shared_md"
        else
            for acct in "${ACCOUNTS[@]}; do
                if [[ -f "$acct/CLAUDE.md" && ! -L "$acct/CLAUDE.md" ]];
                then
                    cp -p "$acct/CLAUDE.md" "$shared_md"; break
                fi
            done
        fi
        [[ -f "$shared_md" ]] || return 0
    fi
    local targets=("$DEFAULT_DIR/CLAUDE.md")
    for acct in "${ACCOUNTS[@]}; do targets+=("$acct/CLAUDE.md");
    done
    for tgt in "${targets[@]}; do
        [[ -d "$(dirname "$tgt")" ]] || continue

```

```

    already_linked_to_shared "$tgt" "$shared_md" && continue
    backup_and_remove "$tgt"
    ln -s "$shared_md" "$tgt"
done
}

rollback() {
    cma_log "rollback: restoring .preunify.* backups"
    local root
    local roots=("$DEFAULT_DIR" "$DEFAULT_DIR/plugins"
        "$SHARED_DIR")
    for root in "${ACCOUNTS[@]};" ; do roots+=("$root"); done
    for root in "${roots[@]};" ; do
        [[ -d "$root" ]] || continue
        while IFS= read -r -d '' bk; do
            local orig="${bk%.preunify.*}"
            [[ -L "$orig" ]] && rm -f "$orig"
            [[ -e "$orig" ]] || { mv "$bk" "$orig"; cma_log "restored
                $orig"; }
        done << (find "$root" -maxdepth 1 -name '*.preunify.*'
            -print0 2>/dev/null)
    done
    if [[ -d "$SHARED_DIR" ]]; then
        mv "$SHARED_DIR" "${SHARED_DIR}.removed.${ts}"
        cma_log "moved $SHARED_DIR aside"
    fi
}

usage() {
    cat <<EOF
Usage: $(basename "$0") [--rollback] [account-dir ...]

Without args, auto-detects all ~/${ACCOUNT_PREFIX}* directories.
With --rollback, restores .preunify.* backups and removes the
    shared store.

Env: SHARED_DIR=$SHARED_DIR DEFAULT_DIR=$DEFAULT_DIR
EOF
}

# === main ===

ACCOUNTS=()
DO_ROLLBACK=0
while (( $# )); do
    case "$1" in
        -h|--help) usage; exit 0 ;;
        --rollback) DO_ROLLBACK=1; shift ;;
        --) shift; while (( $# )); do ACCOUNTS+=("$1"); shift; done ;;
        -*) cma_die "unknown flag: $1" ;;
        *) ACCOUNTS+=("$1"); shift ;;
    esac
done

```

```

done

if (( ${#ACCOUNTS[@]} == 0 )); then
    mapfile -t ACCOUNTS < <(cma_detect_accounts)
fi

(( ${#ACCOUNTS[@]} >= 1 )) || cma_die "no account dirs given and
    none auto-detected at ~/${ACCOUNT_PREFIX}*"

cma_log "accounts: ${ACCOUNTS[*]}"
cma_log "shared:   $SHARED_DIR"

if (( DO_ROLLBACK )); then rollback; exit 0; fi

mkdir -p "$SHARED_DIR"
absorb_default_plugins

for item in "${SHARED_ITEMS[@]}; do
    case "$item" in
        history.jsonl)    merge_history_jsonl ;;
        settings.json)   merge_settings_json ;;
        stats-cache.json) merge_file_into_shared "$item" ;;
        *)                merge_dir_into_shared "$item" ;;
    esac
    [[ "$item" == "plugins" ]] && rewrite_plugin_paths
    link_to_shared "$item"
    cma_log "ok: $item"
done

link_default_plugin_subdirs
sync_claude_md

cma_log "done. shared: $SHARED_DIR"
cma_log "private per-account: ${PRIVATE_ITEMS[*]}"

```

8.3 claude-add-account.sh

```

#!/usr/bin/env bash
# claude-add-account.sh – Add a new Claude Code account to the
# multi-account
# setup. Creates the per-account config directory, links every
# shared item
# to the shared store, and registers a shell alias so the new
# account is
# invocable as e.g. `claude3`.
#
# Modes:
# * Interactive (default): prompts for alias name and config dir
#   name,
#   showing defaults that follow the existing pattern.
# * Non-interactive: pass --alias NAME and optionally --dir
#   PATH, e.g.

```

```

#     `claude-add-account.sh --alias work --dir ~/.claude-work`.
#
# After this script runs you still need to authenticate the new
# account
# with Anthropic – the script prints the exact command for that.

set -euo pipefail

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
# shellcheck source=lib.sh
source "$LIB_DIR/lib.sh"

ALIAS_NAME=""
CONFIG_DIR=""
NONINTERACTIVE=0

usage() {
    cat <<EOF
Usage: $(basename "$0") [--alias NAME] [--dir PATH] [--yes]

    --alias NAME    Shell alias to create (e.g. claude3 or work).
                    Default:
                        next free claudeN.
    --dir    PATH    Config directory for the new account. Default:
                    ~/.${ACCOUNT_PREFIX}<alias>.
    --yes           Skip prompts and use defaults / passed values.
EOF
}

while (( $# )); do
    case "$1" in
        -h|--help) usage; exit 0 ;;
        --alias)    ALIAS_NAME="$2"; shift 2 ;;
        --dir)      CONFIG_DIR="$2"; shift 2 ;;
        --yes|-y)   NONINTERACTIVE=1; shift ;;
        *)          cma_die "unknown arg: $1" ;;
    esac
done

# Prompt with a default value, return whatever the user types (or
# default).
prompt() {
    local label="$1" default="$2" answer
    if (( NONINTERACTIVE )); then echo "$default"; return; fi
    read -r -p "$label [$default]: " answer < /dev/tty
    echo "${answer:-$default}"
}

[[ -n "$ALIAS_NAME" ]] || ALIAS_NAME="$(cma_suggest_alias)"
ALIAS_NAME="$(prompt "Alias name" "$ALIAS_NAME")"
cma_validate_alias "$ALIAS_NAME"

```



```

# Reject if alias already exists in the alias file.
if cma_existing_aliases | grep -qx "$ALIAS_NAME"; then
    cma_die "alias '$ALIAS_NAME' already exists in $ALIAS_FILE"
fi

[[ -n "$CONFIG_DIR" ]] || CONFIG_DIR="$HOME/${ACCOUNT_PREFIX}${ALIAS_NAME}"
CONFIG_DIR="$(prompt "Config directory" "$CONFIG_DIR")"
case "$CONFIG_DIR" in /*) ;; *) CONFIG_DIR="$HOME/$CONFIG_DIR" ;;
esac

[[ -d "$CONFIG_DIR" ]] && cma_die "config dir already exists:
    $CONFIG_DIR (refusing to overwrite)"

mkdir -p "$CONFIG_DIR"
cma_log "created $CONFIG_DIR"

# Symlink every shared item into the new account dir. We mirror
# the same
# list that claude-unify uses so a brand-new account starts in
# lockstep
# with the others without re-running the full merge.
SHARED_ITEMS=(
    projects todos tasks plans file-history paste-cache shell-
    snapshots
    session-env telemetry sessions backups cache plugins
    stats-cache.json history.jsonl settings.json CLAUDE.md
)

mkdir -p "$SHARED_DIR"
for item in "${SHARED_ITEMS[@]}; do
    src="$SHARED_DIR/$item"
    tgt="$CONFIG_DIR/$item"
    # If the shared item doesn't yet exist, create an empty
    # placeholder so

    # the symlink isn't dangling. Directories are mkdir'd,
    # files are touched.
    if [[ ! -e "$src" ]]; then
        case "$item" in
            *.json|*.jsonl|*.md) : > "$src" ;;
            *) mkdir -p "$src" ;;
        esac
    fi
    ln -s "$src" "$tgt"
done
cma_log "linked ${#SHARED_ITEMS[@]} shared items into $CONFIG_DIR"

# Add the alias. lib.sh handles dedupe, rc-file sourcing, etc.
cma_write_alias "$ALIAS_NAME" "$CONFIG_DIR"
cma_log "registered alias: $ALIAS_NAME -> $CONFIG_DIR"

cat <<EOF

```

```
[done] new account: $ALIAS_NAME
```

Next steps:

1. Reload your shell (or run: `source $ALIAS_FILE`).
2. Authenticate the new account:
`$ALIAS_NAME /login`
3. After login, run any project – history, memory, plugins, and settings will already match your other accounts via `$SHARED_DIR`.

To remove this account later:

```
$LIB_DIR/claude-remove-account.sh --alias $ALIAS_NAME
```

EOF

8.4 claude-remove-account.sh

```
#!/usr/bin/env bash
# claude-remove-account.sh – Remove a Claude Code account from the
# multi-account setup. Drops the alias from the managed alias file
# and
# (optionally) deletes or archives the per-account config
# directory.
#
# The shared store is left untouched – only this one account's
# symlinks
# and credentials are removed. Other accounts continue using
# shared.

set -euo pipefail

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
# shellcheck source=lib.sh
source "$LIB_DIR/lib.sh"

ALIAS_NAME=""
DELETE_DIR=0
ARCHIVE_DIR=1
NONINTERACTIVE=0

usage() {
    cat <<EOF
Usage: $(basename "$0") --alias NAME [--delete | --archive] [--
yes]

    --alias NAME    Required. Alias to remove (e.g. claude3).
    --delete        Permanently delete the per-account config
                    directory.
    --archive        Move the per-account dir to
                    <dir>.removed.<timestamp>
                    instead of deleting (default).
    --yes            Skip confirmation prompts.
EOF
}
```

```

EOF
}

while (( $# )); do
    case "$1" in
        -h|--help) usage; exit 0 ;;
        --alias) ALIAS_NAME="$2"; shift 2 ;;
        --delete) DELETE_DIR=1; ARCHIVE_DIR=0; shift ;;
        --archive) ARCHIVE_DIR=1; DELETE_DIR=0; shift ;;
        --yes|-y) NONINTERACTIVE=1; shift ;;
        *) cma_die "unknown arg: $1" ;;
    esac
done

[[ -n "$ALIAS_NAME" ]] || { usage; exit 2; }
cma_validate_alias "$ALIAS_NAME"

# Resolve the config dir by inspecting the existing alias line.
CONFIG_DIR=""
if [[ -f "$ALIAS_FILE" ]]; then
    CONFIG_DIR="$(awk -v a="$ALIAS_NAME" '
        $0 ~ "^alias[:space:]+a=" {
            match($0, /CLAUDE_CONFIG_DIR=([^\s]+)/, m);
            print m[1]; exit
        }' "$ALIAS_FILE")"
fi
[[ -n "$CONFIG_DIR" ]] || cma_die "alias '$ALIAS_NAME' not found
in $ALIAS_FILE"

cma_log "removing alias '$ALIAS_NAME' (config dir: $CONFIG_DIR)"

if (( ! NONINTERACTIVE )); then
    read -r -p "Proceed? [y/N] " ans < /dev/tty
    [[ "$ans" =~ ^[Yy]$ ]] || cma_die "aborted"
fi

cma_remove_alias "$ALIAS_NAME"
cma_log "alias removed from $ALIAS_FILE"

if [[ -d "$CONFIG_DIR" ]]; then
    if (( DELETE_DIR )); then
        rm -rf -- "$CONFIG_DIR"
        cma_log "deleted $CONFIG_DIR"
    else
        mv -- "$CONFIG_DIR" "${CONFIG_DIR}.removed.${date +%Y%m%d%H%M%S}"
        cma_log "archived $CONFIG_DIR -> ${CONFIG_DIR}.removed.*"
    fi
fi

cat <<EOF

```

```
[done] account '$ALIAS_NAME' removed.
```

Notes:

- * Shared store \$SHARED_DIR is untouched – other accounts still work.
- * Reload your shell so the alias change takes effect.

EOF

8.5 claude-list-accounts.sh

```
#!/usr/bin/env bash
# claude-list-accounts.sh – Print a status summary of every
# detected
# Claude Code account: which alias maps to it, whether credentials
# are
# present, and whether its shared-state symlinks are intact.

set -euo pipefail

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
# shellcheck source=lib.sh
source "$LIB_DIR/lib.sh"

# Build alias -> dir mapping from the managed alias file.
declare -A ALIAS_OF_DIR
if [[ -f "$ALIAS_FILE" ]]; then
    while IFS= read -r line; do
        [[ "$line" =~ ^alias[[:space:]]+([a-zA-Z0-9_-]+)
            =.*CLAUDE_CONFIG_DIR=([^[:space:]]+) ]] || continue
        ALIAS_OF_DIR["${BASH_REMATCH[2]}"]="${BASH_REMATCH[1]}"
    done < "$ALIAS_FILE"
fi

# Items we expect to be symlinks pointing into $SHARED_DIR.
CHECK_LINKS=(projects history.jsonl settings.json plugins todos
    CLAUDE.md)

printf '%-12s  %-45s  %-6s  %-5s  %s\n' \
    "ALIAS" "CONFIG DIR" "CREDS" "LINKS" "NOTES"

mapfile -t accounts < <(cma_detect_accounts)
for dir in "${accounts[@]}; do
    alias_name="${ALIAS_OF_DIR[$dir]:--}"
    if [[ -f "$dir/.credentials.json" ]]; then creds="yes"; else
        creds="no"; fi
    ok=0 total=0 missing=()
    for item in "${CHECK_LINKS[@]}; do
        total=$((total+1))
        if [[ -L "$dir/$item" ]] \
            && [[ "$(readlink -f "$dir/$item")" == "$(readlink -f
                "$SHARED_DIR/$item" 2>/dev/null || echo "$SHARED_DIR/
                $item")" ]]; then
```

```

        ok=$((ok+1))
    else
        missing+=("$item")
    fi
done
notes=""
(( ${#missing[@]} )) && notes="not linked: ${missing[*]}"
printf '%-12s  %-45s  %-6s  %-5s  %s\n' \
    "$alias_name" "$dir" "$creds" "$ok/$total" "$notes"
done

echo
echo "Shared store: $SHARED_DIR"
[[ -d "$SHARED_DIR" ]] || echo " (does not exist – run claude-
unify.sh)"
echo "Alias file: $ALIAS_FILE"
[[ -f "$ALIAS_FILE" ]] || echo " (does not exist – run
install.sh)"

```

8.6 claude-rollback.sh

```

#!/usr/bin/env bash
# claude-rollback.sh – Convenience wrapper that calls claude-
unify.sh with
# --rollback. Restores every .preunify.<timestamp> backup created
by the
# unification run and archives the shared store out of the way.
#
# After this you can re-run install.sh to start over from scratch.

set -euo pipefail

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
exec "$LIB_DIR/claude-unify.sh" --rollback "$@"

```

8.7 claude-export-docs.sh

```

#!/usr/bin/env bash
# claude-export-docs.sh – Convert the multi-account markdown doc
to
# matching .html and .pdf siblings. Re-runnable; overwrites
outputs.
#
# Order of preference for PDF: pandoc + weasyprint > pandoc +
wkhtmltopdf
# > chromium --headless print-to-pdf > weasyprint on the generated
HTML.
# HTML always uses pandoc with a self-contained option so the file
is
# portable without external CSS/JS.

set -euo pipefail

```

```

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
# shellcheck source=lib.sh
source "$LIB_DIR/lib.sh"

DOC_DIR="${DOC_DIR:-$HOME/Documents}"
MD_FILE="${MD_FILE:-$DOC_DIR/Claude_Multi_Account_Fine_Tuning.md}"
HTML_FILE="${MD_FILE%.md}.html"
PDF_FILE="${MD_FILE%.md}.pdf"

[[ -f "$MD_FILE" ]] || cma_die "markdown not found: $MD_FILE"
cma_require pandoc

# --- Preprocess: expand `<!-- INCLUDE: relative/path -->` markers
# ---
# These markers, placed inside a fenced code block in the source
# markdown,
# get replaced with the contents of the referenced file (relative
# to the
# directory containing the markdown). That keeps the .md small
# while the
# generated HTML/PDF stays self-contained.
PROCESSED_MD="$(mktemp --suffix=.md)"
trap 'rm -f "$PROCESSED_MD"' EXIT

awk -v base="$(dirname "$MD_FILE")" '
  /<!-- INCLUDE: / {
    match($0, /INCLUDE: ([^ ]+) -->/, m)
    if (m[1] != "") {
      path = m[1]
      if (path !~ /^\/\//) path = base "/" path
      while ((getline line < path) > 0) print line
      close(path)
      next
    }
  }
  { print }
' "$MD_FILE" > "$PROCESSED_MD"

# --- HTML ---
# --standalone makes a complete HTML doc (head/body), --embed-
# resources
# inlines CSS/JS/images so the file is self-contained.
cma_log "rendering HTML -> $HTML_FILE"
pandoc \
  --from=gfm+yaml_metadata_block \
  --to=html5 \
  --standalone \
  --embed-resources \
  --toc --toc-depth=3 \
  --highlight-style=tango \
  --metadata title="Claude Multi-Account Fine Tuning" \
  -o "$HTML_FILE" \

```

```

"$PROCESSED_MD"

# --- PDF ---
cma_log "rendering PDF -> $PDF_FILE"

render_pdf_via_pandoc_engine() {
    local engine="$1"
    command -v "$engine" >/dev/null 2>&1 || return 1
    pandoc \
        --from=gfm+yaml_metadata_block \
        --pdf-engine="$engine" \
        --toc --toc-depth=3 \
        --highlight-style=tango \
        -V geometry:margin=0.9in \
        -V mainfont="DejaVu Serif" \
        -V monofont="DejaVu Sans Mono" \
        --metadata title="Claude Multi-Account Fine Tuning" \
        -o "$PDF_FILE" \
        "$PROCESSED_MD"
}

render_pdf_via_chromium() {
    local engine
    for engine in chromium chromium-browser google-chrome chrome; do
        if command -v "$engine" >/dev/null 2>&1; then
            # Use the freshly generated HTML so styling matches.
            "$engine" --headless --no-sandbox --disable-gpu \
                --print-to-pdf="$PDF_FILE" \
                --print-to-pdf-no-header \
                "file://$HTML_FILE" >/dev/null 2>&1
            return $?
        fi
    done
    return 1
}

render_pdf_via_weasyprint_on_html() {
    command -v weasyprint >/dev/null 2>&1 || return 1
    weasyprint "$HTML_FILE" "$PDF_FILE"
}

# Try engines in order; the first that succeeds wins.
if render_pdf_via_pandoc_engine weasyprint 2>/dev/null; then
    cma_log "pdf engine: weasyprint (via pandoc)"
elif render_pdf_via_pandoc_engine wkhtmltopdf 2>/dev/null; then
    cma_log "pdf engine: wkhtmltopdf (via pandoc)"
elif render_pdf_via_weasyprint_on_html 2>/dev/null; then
    cma_log "pdf engine: weasyprint on html"
elif render_pdf_via_chromium 2>/dev/null; then
    cma_log "pdf engine: chromium headless"
else

```

```

cma_die "no usable PDF engine. Install one of: weasyprint,
        wkhtmltopdf, chromium."
fi

ls -lh "$HTML_FILE" "$PDF_FILE"

```

8.8 install.sh

```

#!/usr/bin/env bash
# install.sh – One-shot bootstrap for the Claude multi-account
# toolkit.
# Idempotent; safe to re-run after pulling updates.
#
# What it does, in order:
# 1. Detect tooling (jq, rsync, awk, pandoc) and fail fast if
#    missing.
# 2. Symlink every script into ~/.local/bin (creates the dir if
#    needed).
# 3. Ensure ~/.local/bin is on PATH for future shells.
# 4. Create the managed alias file at $ALIAS_FILE and source it
#    from
#    ~/.bashrc and ~/.zshrc (whichever exist).
# 5. Run claude-unify.sh to merge any existing account dirs.
# 6. Run claude-export-docs.sh to refresh the PDF/HTML alongside
#    the
#    markdown doc.

set -euo pipefail

LIB_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
# shellcheck source=lib.sh
source "$LIB_DIR/lib.sh"

cma_log "platform: $(cma_os)"

# 1. Required tooling. pandoc is needed only for doc export; the
#    rest are
# load-bearing for unify/add-account.
for t in rsync jq awk; do cma_require "$t"; done
command -v pandoc >/dev/null 2>&1 || cma_warn "pandoc not found –
        PDF/HTML export will be skipped"

# 2. Symlink the scripts onto PATH.
BIN_DIR="${BIN_DIR:-$HOME/.local/bin}"
mkdir -p "$BIN_DIR"
for f in "$LIB_DIR"/claude-*.sh; do
    name="$(basename "$f" .sh)"
    link="$BIN_DIR/$name"
    if [[ -L "$link" || -e "$link" ]]; then
        if [[ "$(readlink -f "$link" 2>/dev/null)" != "$(readlink -f
            "$f")" ]]; then
            mv "$link" "${link}.preunify.${date +%Y%m%d%H%M%S}"
            ln -s "$f" "$link"

```



```

        cma_log "linked $link -> $f"
    fi
else
    ln -s "$f" "$link"
    cma_log "linked $link -> $f"
fi
done

# 3. Make sure ~/.local/bin is on PATH for new shells. We add
#     to .bashrc
# and .zshrc once; existing shells need a manual reload.
PATH_LINE='export PATH="$HOME/.local/bin:$PATH"'
for rc in "$HOME/.bashrc" "$HOME/.zshrc"; do
    [[ -f "$rc" ]] || continue
    if ! grep -F -q "$PATH_LINE" "$rc"; then
        printf '\n# Claude multi-account: ensure ~/.local/bin is on
            PATH\n%s\n' "$PATH_LINE" >> "$rc"
        cma_log "added PATH line to $rc"
    fi
done

# 4. Alias file + rc sourcing.
cma_ensure_alias_file

# Re-register any pre-existing aliases from $HOME/.bashrc into the
#     managed
# alias file, then comment out the originals so they're not
#     duplicated.
migrate_inline_aliases() {
    local rc="$1"
    [[ -f "$rc" ]] || return 0
    local tmp; tmp="$(mktemp)"
    local changed=0
    while IFS= read -r line; do
        if [[ "$line" =~ ^alias[[:space:]]+(claude[0-9a-zA-Z_-]
            +)=.*CLAUDE_CONFIG_DIR=([[:space:]]\")+ ]]; then
            local a="${BASH_REMATCH[1]}" d="${BASH_REMATCH[2]}"
            d="${d//\"/}"
            d="${d/#\$HOME/$HOME}"
            d="${d/#~/$HOME}"
            cma_write_alias "$a" "$d"
            printf '# migrated to %s: %s\n' "$ALIAS_FILE" "$line" >>
                "$tmp"
            changed=1
        else
            printf '%s\n' "$line" >> "$tmp"
        fi
    done < "$rc"
    if (( changed )); then
        cp -p "$rc" "${rc}.preunify.${date +%Y%m%d%H%M%S}"
        mv "$tmp" "$rc"
        cma_log "migrated inline claude* aliases in $rc ->
            $ALIAS_FILE"
    fi
}

```

```

else
    rm -f "$tmp"
fi
}
migrate_inline_aliases "$HOME/.bashrc"
migrate_inline_aliases "$HOME/.zshrc"

# 5. Unify whatever accounts exist now.
if (( $(cma_detect_accounts | wc -l) > 0 )); then
    cma_log "running claude-unify.sh"
    "$LIB_DIR/claude-unify.sh"
else
    cma_warn "no ~/${ACCOUNT_PREFIX}* dirs detected – add one with
    claude-add-account.sh"
fi

# 6. Refresh docs if pandoc is available.
if command -v pandoc >/dev/null 2>&1 && [[ -f "$HOME/Documents/
    Claude_Multi_Account_Fine_Tuning.md" ]]; then
    cma_log "running claude-export-docs.sh"
    "$LIB_DIR/claude-export-docs.sh" || cma_warn "doc export failed
    (continuing)"
fi

cat <<EOF

[done] claude-multi-account installed.

    Scripts on PATH:  $BIN_DIR/claude-{unify,add-account,remove-
        account,list-accounts,rollback,export-docs}
    Alias file:       $ALIAS_FILE
    Shared store:     $SHARED_DIR

Open a new shell (or run: source
    $ALIAS_FILE) so the aliases load, then:
    claude-list-accounts      # see what's wired up
    claude-add-account        # add another account
    interactively

EOF

```

8.9 tests/lib/assert.sh

```

#!/usr/bin/env bash
# assert.sh – small assertion helpers for the test suite. Each
# helper
# emits `[PASS]` or `[FAIL]` lines and increments counters in the
# caller.
# A test file should source this once and then call the helpers
# inside
# `it` blocks.

TESTS_PASSED="${TESTS_PASSED:-0}"
TESTS_FAILED="${TESTS_FAILED:-0}"

```

```

TESTS_FAILURES=()
CURRENT_TEST="${CURRENT_TEST:- (unknown)}"

# Mark the start of an individual test case. Use as: it "merges
# history.jsonl"
it() {
    CURRENT_TEST="$*"
    printf '\n • %s\n' "$CURRENT_TEST"
}

_pass() {
    TESTS_PASSED=$((TESTS_PASSED + 1))
    printf '\033[32m[PASS]\033[0m %s\n' "$1"
}

_fail() {
    TESTS_FAILED=$((TESTS_FAILED + 1))
    TESTS_FAILURES+=("$CURRENT_TEST :: $1")
    printf '\033[31m[FAIL]\033[0m %s\n' "$1"
    [[ -n "${2:-}" ]] && printf '%s\n' "$2"
}

assert_eq() {
    local expected="$1" actual="$2" msg="${3:-equality}"
    if [[ "$expected" == "$actual" ]]; then _pass "$msg ($expected)"
    else _fail "$msg" "want=$expected got=$actual"; fi
}

assert_file() {
    local path="$1" msg="${2:-file exists}"
    if [[ -f "$path" ]]; then _pass "$msg: $path"
    else _fail "$msg" "missing: $path"; fi
}

assert_dir() {
    local path="$1" msg="${2:-dir exists}"
    if [[ -d "$path" ]]; then _pass "$msg: $path"
    else _fail "$msg" "missing dir: $path"; fi
}

assert_symlink_to() {
    local link="$1" expected="$2" msg="${3:-symlink target}"
    if [[ ! -L "$link" ]]; then
        _fail "$msg" "not a symlink: $link"
        return
    fi
    local actual; actual="$(readlink -f "$link")"
    local want; want="$(readlink -f "$expected" 2>/dev/null ||
        echo "$expected")"
    if [[ "$actual" == "$want" ]]; then _pass "$msg: $link ->
        $expected"
    else _fail "$msg" "want=$want got=$actual"; fi
}

```

```

}

assert_not_symlink() {
    local p="$1" msg="${2:-not a symlink}"
    if [[ -L "$p" ]]; then _fail "$msg" "is a symlink: $p"
    else _pass "$msg: $p"; fi
}

assert_file_contains() {
    local path="$1" needle="$2" msg="${3:-file contains}"
    if grep -F -q -- "$needle" "$path" 2>/dev/null; then _pass
        "$msg: '$needle' in $(basename "$path")"
    else _fail "$msg" "'$needle' not found in $path"; fi
}

assert_file_not_contains() {
    local path="$1" needle="$2" msg="${3:-file lacks}"
    if grep -F -q -- "$needle" "$path" 2>/dev/null; then _fail
        "$msg" "'$needle' found in $path"
    else _pass "$msg: '$needle' absent from $(basename "$path")"; fi
}

assert_jq() {
    local path="$1" expr="$2" expected="$3" msg="${4:-jq check}"
    local actual; actual="$(jq -r "$expr" "$path" 2>/dev/null ||
        echo '<jq-error>')"
    if [[ "$actual" == "$expected" ]]; then _pass "$msg ($expr =
        $expected)"
    else _fail "$msg" "expr=$expr want=$expected got=$actual"; fi
}

assert_lines() {
    local path="$1" expected="$2" msg="${3:-line count}"
    local actual; actual="$(wc -l < "$path" 2>/dev/null || echo 0)"
    actual="${actual// /}"
    if [[ "$actual" == "$expected" ]]; then _pass "$msg ($expected
        lines)"
    else _fail "$msg" "want=$expected got=$actual lines in $path";
        fi
}

assert_exit() {
    local expected="$1"; shift
    local out; out="$("$@" 2>&1)"; local rc=$?
    if (( rc == expected )); then _pass "exit $expected for: $*"
    else _fail "exit code mismatch" "want=$expected got=$rc cmd=$*
        out=$out"; fi
}

summary() {
    echo
    if (( TESTS_FAILED == 0 )); then

```

```

    printf '\033[32m✓ %d passed, 0 failed\033[0m\n'
        "$TESTS_PASSED"
    return 0
fi
printf '\033[31m✗ %d failed, %d passed\033[0m\n'
    "$TESTS_FAILED" "$TESTS_PASSED"
printf '\nFailures:\n'
for f in "${TESTS_FAILURES[@]}; do printf '  - %s\n' "$f"; done
return 1
}

```

8.10 tests/lib/sandbox.sh

```

#!/usr/bin/env bash
# sandbox.sh – utilities for running scripts against a temp HOME
#               so the
#               real ~/.claude state is never touched.
#
# Usage from a test file:
#
#   source "$TESTS_DIR/lib/sandbox.sh"
#   make_sandbox          # creates $SANDBOX_HOME, exports HOME,
#                           etc.
#   make_account acct1    # creates a fake ~/.claude-acct1 inside
#                           sandbox
#   make_account acct2 --plugins
#   run_unify             # runs claude-unify.sh against the
#                           sandbox
#   cleanup_sandbox       # rm -rf the sandbox (registered with
#                           trap)

```

```
SCRIPTS_DIR="${SCRIPTS_DIR:-$HOME/Documents/scripts}"
```

```

make_sandbox() {
    SANDBOX_HOME="$(mktemp -d -t cma-test.XXXXXXXXX)"
    # Override every env var the toolkit reads from. Pointing HOME
    #   at the
    #   sandbox is the main switch; the other vars only matter if a
    #   caller
    #   tests defaults explicitly.
    export HOME="$SANDBOX_HOME"
    export SHARED_DIR="$SANDBOX_HOME/.claude-shared"
    export ALIAS_FILE="$SANDBOX_HOME/.local/share/claude-multi-
        account/aliases.sh"
    export DEFAULT_DIR="$SANDBOX_HOME/.claude"
    export ACCOUNT_PREFIX=".claude-"
    export CLAUDE_BIN="/usr/bin/true" # dummy; we never actually
        launch claude
    mkdir -p "$DEFAULT_DIR" "$SANDBOX_HOME/.local/bin"
    trap 'cleanup_sandbox' EXIT
}

```

```
cleanup_sandbox() {
```

```

[[ -n "${SANDBOX_HOME:-}" ]] && -d "$SANDBOX_HOME" ]] || return 0
# Safety net: only delete if the path matches what mktemp would
# have
# produced (cma-test.* somewhere under a tempdir root).
# Different
# systems put mktemp output in different places (/tmp on most
# Linux,
# /tmp/.private/<user>/ on this host, /var/folders/... on
# macOS) so
# we anchor on the cma-test. prefix rather than the parent.
case "$(basename "$SANDBOX_HOME")" in
  cma-test.*) rm -rf -- "$SANDBOX_HOME" ;;
  *) echo "refusing to rm sandbox at unexpected path:
    $SANDBOX_HOME" >&2 ;;
esac
}

# make_account NAME [--plugins] [--settings JSON] [--history
# "line1|line2"] [--memory KEY:VALUE...]
make_account() {
  local name="$1" with_plugins=0 settings_json="" history_lines=()
  shift
  local memory_entries=() todo_entries=()
  while (( $# )); do
    case "$1" in
      --plugins) with_plugins=1; shift ;;
      --settings) settings_json="$2"; shift 2 ;;
      --history) IFS='|' read -r -a history_lines <<< "$2"; shift
        2 ;;
      --memory) memory_entries+=("$2"); shift 2 ;;
      --todo) todo_entries+=("$2"); shift 2 ;;
      *) echo "make_account: unknown arg $1" >&2; return 1 ;;
    esac
  done

  local dir="$HOME/${ACCOUNT_PREFIX}${name}"
  mkdir -p "$dir/projects/-home-test/memory" "$dir/todos" "$dir/
    tasks" \
    "$dir/plans" "$dir/file-history" "$dir/paste-cache" \
    "$dir/shell-snapshots" "$dir/session-env" "$dir/
    telemetry" \
    "$dir/sessions" "$dir/backups" "$dir/cache" "$dir/
    plugins"

  # Fake credentials so list-accounts reports CREDs:yes.
  printf '{"account":"%s"}\n' "$name" > "$dir/.credentials.json"
  printf '{"name":"%s"}\n' "$name" > "$dir/.claude.json"

  # A unique transcript so the union-merge step has something to
  # merge.
  printf '{"role":"user","msg":"hi from %s"}\n' "$name" \
    > "$dir/projects/-home-test/$(uuidgen 2>/dev/null || echo
    "session-${name}").jsonl"

```

```

# Memory + todo content if requested.
local m
for m in "${memory_entries[@]}; do
    local k="${m%*:}" v="${m#*:}"
    printf -- '%s\n' "$v" > "$dir/projects/-home-test/memory/${k}.md"
done
local t
for t in "${todo_entries[@]}; do
    printf '{"subject":"%s"}\n' "$t" > "$dir/todos/${uuidgen 2>/dev/null || echo "todo-${name}-${t}").json"
done

# History lines (used to test concat+dedup later).
if (( ${#history_lines[@]} )); then
    printf '%s\n' "${history_lines[@]}" > "$dir/history.jsonl"
fi

# Settings JSON if specified, else a minimal default.
if [[ -n "$settings_json" ]]; then
    printf '%s\n' "$settings_json" > "$dir/settings.json"
else
    printf '{"enabledPlugins":{"%s-plugin":true}}\n' "$name" >
        "$dir/settings.json"
fi

if (( with_plugins )); then
    mkdir -p "$dir/plugins/cache/test-marketplace/test-plugin/1.0.0"
    mkdir -p "$dir/plugins/marketplaces/test-marketplace"
    cat > "$dir/plugins/installed_plugins.json" <<EOF
{
  "version": 2,
  "plugins": {
    "test-plugin@test-marketplace": [
      {
        "scope": "project",
        "installPath": "$dir/plugins/cache/test-marketplace/test-plugin/1.0.0",
        "version": "1.0.0"
      }
    ]
  }
}
EOF
    cat > "$dir/plugins/known_marketplaces.json" <<EOF
{
  "test-marketplace": {
    "source": {"source": "github", "repo": "example/test"},
    "installLocation": "$dir/plugins/marketplaces/test-marketplace"
  }
}

```

```

}
EOF
fi

    echo "$dir"
}

run_unify() {
    "$SCRIPTS_DIR/claude-unify.sh" "$@"
}

run_add_account() {
    "$SCRIPTS_DIR/claude-add-account.sh" "$@"
}

run_remove_account() {
    "$SCRIPTS_DIR/claude-remove-account.sh" "$@"
}

run_list_accounts() {
    "$SCRIPTS_DIR/claude-list-accounts.sh" "$@"
}

run_rollback() {
    "$SCRIPTS_DIR/claude-rollback.sh" "$@"
}

```

8.11 tests/run-all.sh

```

#!/usr/bin/env bash
# run-all.sh – discover every test_*.sh sibling, run each in its
#               own
# subshell, and tally pass/fail.
#
# Exit code is 0 only if every test file's `summary` returned 0.

set -uo pipefail

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"
export SCRIPTS_DIR

FILES=()
if (( $# )); then
    for arg in "$@"; do FILES+=("$TESTS_DIR/test_${arg}.sh"); done
else
    mapfile -t FILES << (find "$TESTS_DIR" -maxdepth 1 -name
        'test_*.sh' -type f | sort)
fi

PASSED=0 FAILED=0 FAILED_FILES=()

```



```

for f in "${FILES[@]"; do
    [[ -f "$f" ]] || { echo "missing: $f" >&2; FAILED=$((FAILED+1)); FAILED_FILES+=("$f"); continue; }
    printf '\n\033[1m==> %s\033[0m\n' "${basename "$f"}"
    if bash "$f"; then
        PASSED=$((PASSED + 1))
    else
        FAILED=$((FAILED + 1))
        FAILED_FILES+=("${basename "$f}")
    fi
done

echo
echo "=====
echo "Test files: $((PASSED + FAILED))    passed: $PASSED
        failed: $FAILED"
if (( FAILED )); then
    echo "Failed files:"
    for f in "${FAILED_FILES[@]"; do echo "    - $f"; done
    exit 1
fi
echo "ALL GREEN"

```

8.12 tests/test_lib.sh

```

#!/usr/bin/env bash
# test_lib.sh – unit tests for the helper functions in lib.sh.
set -uo pipefail

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"

# shellcheck source=lib/assert.sh
source "$TESTS_DIR/lib/assert.sh"
# shellcheck source=lib/sandbox.sh
source "$TESTS_DIR/lib/sandbox.sh"

make_sandbox
# shellcheck source=../lib.sh
source "$SCRIPTS_DIR/lib.sh"
# lib.sh enables `set -e`. The test harness is intentionally
# tolerant of
# non-zero exits (we assert on them), so turn it back off here.
set +e

it "cma_validate_alias accepts well-formed names"
( set -e; cma_validate_alias "claud3" ); assert_eq 0 $? "claud3"
( set -e; cma_validate_alias "work_acct-1" ); assert_eq 0 $?
    "work_acct-1"

it "cma_validate_alias rejects invalid names"

```

```

( cma_validate_alias "3claude" >/dev/null 2>&1 ); assert_eq 1 $?
    "rejects digit-leading"
( cma_validate_alias "bad name" >/dev/null 2>&1 ); assert_eq 1 $?
    "rejects spaces"
( cma_validate_alias "" >/dev/null 2>&1 ); assert_eq 1 $?
    "rejects empty"

it "cma_suggest_alias starts at claude1 when nothing exists"
suggestion="$(cma_suggest_alias)"
assert_eq "claude1" "$suggestion" "first suggestion"

it "cma_suggest_alias increments past existing claudeN aliases"
cma_write_alias claude1 "$HOME/.claude-acct1"
cma_write_alias claude2 "$HOME/.claude-acct2"
cma_write_alias claude5 "$HOME/.claude-acct5"    # gap
suggestion="$(cma_suggest_alias)"
assert_eq "claude6" "$suggestion" "skips past highest, not count"

it "cma_write_alias is idempotent – rewriting same alias doesn't
    duplicate"
cma_write_alias claude1 "$HOME/.claude-acct1"    # second write
count="$(grep -c '^alias claude1=' "$ALIAS_FILE")"
assert_eq "1" "$count" "one alias line for claude1"

it "cma_remove_alias removes the line"
cma_remove_alias claude5
assert_file_not_contains "$ALIAS_FILE" "alias claude5=" "claude5
    removed"

it "cma_ensure_alias_file sources from .bashrc"
touch "$HOME/.bashrc"
rm -f "$ALIAS_FILE"
cma_ensure_alias_file
assert_file "$ALIAS_FILE" "alias file created"
assert_file_contains "$HOME/.bashrc" "source \"$ALIAS_FILE\""
    ".bashrc gets source line"

it "cma_detect_accounts skips the shared store"
mkdir -p "$HOME/.claude-shared" "$HOME/.claude-acct1"
mapfile -t found < <(cma_detect_accounts)
joined="${found[*]}"
[[ "$joined" == *".claude-acct1"* ]]; assert_eq 0 $?
    "finds .claude-acct1"
[[ "$joined" != *".claude-shared"* ]]; assert_eq 0 $?
    "excludes .claude-shared"

summary

```

8.13 tests/test_unify.sh

```

#!/usr/bin/env bash
# test_unify.sh – exercises claude-unify.sh:

```

```

# * basic merge of two accounts: all shared items become
#     symlinks
# * settings.json enabledPlugins is a strict union
# * history.jsonl is concatenated and de-duplicated
# * memory files survive merge (newest account wins conflicts)
# * plugin manifest installPath is rewritten to the shared store
# * N>2 accounts also work
# * a second unify run produces the same state (idempotent)
# * --rollback restores .preunify.* backups

set -uo pipefail

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"

source "$TESTS_DIR/lib/assert.sh"
source "$TESTS_DIR/lib/sandbox.sh"

make_sandbox

# Two fixture accounts with overlapping and unique data.
acct1="$(make_account acct1 --plugins \
  --settings '{"enabledPlugins":
    {"a":true,"b":true},"effortLevel":"high"}' \
  --history 'cmd1|cmd2|shared-line' \
  --memory 'user_role:role from acct1' \
  --todo todo-a-1)"
acct2="$(make_account acct2 --plugins \
  --settings '{"enabledPlugins":
    {"b":true,"c":true},"skipDangerousModePermissionPrompt":true}' \
  \
  --history 'cmd3|shared-line|cmd4' \
  --memory 'user_role:role from acct2 (NEWER)' \
  --memory 'project_notes:notes from acct2 only' \
  --todo todo-b-1)"

# Make sure acct2 is the most recently touched so the unifier's
# "last
# account wins on conflicts" rule applies to it.
touch "$acct2/projects/-home-test/memory/user_role.md"

run_unify >/dev/null 2>&1
rc=$?
assert_eq 0 "$rc" "claude-unify exits 0"

it "every shared item becomes a symlink into the shared store"
for item in projects todos tasks plans file-history paste-cache
  plugins \
    shell-snapshots session-env telemetry sessions backups
  cache \
    stats-cache.json history.jsonl settings.json; do
  case "$item" in
    stats-cache.json)

```

```

    # Only fixtures with this file get the symlink; skip if
    absent.
    [[ -e "$SHARED_DIR/$item" ]] || continue ;;
esac
assert_symlink_to "$acct1/$item" "$SHARED_DIR/$item" "$acct1
$item linked"
assert_symlink_to "$acct2/$item" "$SHARED_DIR/$item" "$acct2
$item linked"
done

it "settings.json enabledPlugins is the union {a,b,c}"
assert_jq "$SHARED_DIR/settings.json" '.enabledPlugins | keys |
sort | join(",")' "a,b,c" "union"
assert_jq "$SHARED_DIR/settings.json" '.effortLevel' "high"
"effortLevel preserved"
assert_jq "$SHARED_DIR/settings.json"
'.skipDangerousModePermissionPrompt' "true" "permission
flag preserved"

it "history.jsonl contains every unique line, no duplicates"
expected_lines=5
actual_lines="$(sort -u "$SHARED_DIR/history.jsonl" | wc -l | tr
-d ' ')"
assert_eq "$expected_lines" "$actual_lines" "5 unique lines"
assert_file_contains "$SHARED_DIR/history.jsonl" "cmd1" "cmd1
present"
assert_file_contains "$SHARED_DIR/history.jsonl" "cmd4" "cmd4
present"

it "memory: newest account's version wins on filename conflict"
assert_file_contains "$SHARED_DIR/projects/-home-test/memory/
user_role.md" "NEWER" "acct2 wins"
assert_file "$SHARED_DIR/projects/-home-test/memory/
project_notes.md" "acct2-only memory survives"

it "plugin manifest paths are rewritten to the shared store"
assert_jq "$SHARED_DIR/plugins/installed_plugins.json" \
'.plugins["test-plugin@test-marketplace"][0].installPath' \
"$SHARED_DIR/plugins/cache/test-marketplace/test-plugin/1.0.0" \
"installPath rewritten"
assert_jq "$SHARED_DIR/plugins/known_marketplaces.json" \
'.["test-marketplace"].installLocation' \
"$SHARED_DIR/plugins/marketplaces/test-marketplace" \
"installLocation rewritten"

it "credentials are NOT shared – each account keeps its own"
assert_not_symlink "$acct1/.credentials.json" "acct1 creds local"
assert_not_symlink "$acct2/.credentials.json" "acct2 creds local"
assert_file_contains "$acct1/.credentials.json" "acct1" "acct1
creds intact"
assert_file_contains "$acct2/.credentials.json" "acct2" "acct2
creds intact"

```

```

it "re-running claude-unify is idempotent (settings.json byte-
    stable)"
checksum_before="$(sha256sum "$SHARED_DIR/settings.json" | cut
    -d' ' -f1)"
target_before="$(readlink -f "$acct1/projects")"
run_unify >/dev/null 2>&1
checksum_after="$(sha256sum "$SHARED_DIR/settings.json" | cut -d'
    ' -f1)"
target_after="$(readlink -f "$acct1/projects")"
assert_eq "$checksum_before" "$checksum_after" "settings.json
    unchanged"
assert_eq "$target_before" "$target_after" "symlink target
    unchanged"

it "N>2 accounts: adding a third and re-running unifies it too"
acct3="$(make_account acct3 \
    --settings '{"enabledPlugins":{"d":true}}' \
    --history 'cmd5|cmd1')"
run_unify >/dev/null 2>&1
assert_symlink_to "$acct3/projects" "$SHARED_DIR/projects" "acct3
    projects linked"
assert_jq "$SHARED_DIR/settings.json" '.enabledPlugins.d' "true"
    "acct3 plugin merged"
assert_file_contains "$SHARED_DIR/history.jsonl" "cmd5" "acct3
    history merged"

it "--rollback restores backups and archives the shared store"
run_rollback >/dev/null 2>&1
assert_dir "$acct1/projects" "acct1 projects restored as real dir"
assert_not_symlink "$acct1/projects" "no longer a symlink"
[[ ! -d "$SHARED_DIR" ]]; assert_eq 0 $? "shared store moved
    aside"
shared_archive="$(find "$HOME" -maxdepth 1 -name '.claude-
    shared.removed.*' -type d 2>/dev/null | head -1)"
[[ -n "$shared_archive" ]]; assert_eq 0 $? "archive sibling
    exists"

summary

```

8.14 tests/test_add_remove.sh

```

#!/usr/bin/env bash
# test_add_remove.sh – exercises claude-add-account.sh and
# claude-remove-account.sh end-to-end:
# * add suggests the next free claudeN
# * add creates the dir with the right symlinks
# * add writes a shell-safe alias line
# * add refuses to overwrite an existing dir
# * add accepts a custom alias name
# * remove drops the alias line
# * remove archives the dir by default, --delete removes it

set -uo pipefail

```

```

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"

source "$TESTS_DIR/lib/assert.sh"
source "$TESTS_DIR/lib/sandbox.sh"

make_sandbox

# Start with two existing accounts and a unified shared store.
acct1="$(make_account acct1)"
acct2="$(make_account acct2)"
run_unify >/dev/null 2>&1

it "claude-add-account --yes uses sensible defaults"
run_add_account --alias claude3 --yes >/dev/null 2>&1
rc=$?
assert_eq 0 "$rc" "exit 0"
new_dir="$HOME/.claude-claude3"
assert_dir "$new_dir" "config dir created"
assert_symlink_to "$new_dir/projects" "$SHARED_DIR/projects"
    "projects linked"
assert_symlink_to "$new_dir/settings.json" "$SHARED_DIR/
    settings.json" "settings linked"
assert_symlink_to "$new_dir/CLAUDE.md" "$SHARED_DIR/CLAUDE.md"
    "CLAUDE.md linked"
assert_file_contains "$ALIAS_FILE" "alias claude3=" "alias line
    written"
assert_file_contains "$ALIAS_FILE" "CLAUDE_CONFIG_DIR=$new_dir"
    "alias points at new dir"

it "claude-add-account refuses to overwrite an existing dir"
( run_add_account --alias claude3 --yes >/dev/null 2>&1 )
rc=$?
[[ $rc -ne 0 ]]; assert_eq 0 $? "exits non-zero when dir exists"

it "claude-add-account accepts a custom alias name and custom dir"
custom_dir="$HOME/.claude-mywork"
run_add_account --alias mywork --dir "$custom_dir" --yes >/dev/
    null 2>&1
assert_dir "$custom_dir" "custom dir created"
assert_symlink_to "$custom_dir/projects" "$SHARED_DIR/projects"
    "projects linked"
assert_file_contains "$ALIAS_FILE" "alias mywork=" "custom alias
    written"

it "claude-add-account validates alias names"
( run_add_account --alias "bad name" --yes >/dev/null 2>&1 )
rc=$?
[[ $rc -ne 0 ]]; assert_eq 0 $? "rejects bad alias"

it "claude-remove-account --archive moves the dir aside"
run_remove_account --alias claude3 --archive --yes >/dev/null 2>&1

```

```

rc=$?
assert_eq 0 "$rc" "exit 0"
[[ ! -d "$HOME/.claude-claude3" ]]; assert_eq 0 $? "original dir
gone"
archived="$(find "$HOME" -maxdepth 1 -name '.claude-
claude3.removed.*' -type d 2>/dev/null | head -1)"
[[ -n "$archived" ]]; assert_eq 0 $? "archived sibling exists"
assert_file_not_contains "$ALIAS_FILE" "alias claude3=" "alias
line removed"

it "claude-remove-account --delete actually deletes"
run_remove_account --alias mywork --delete --yes >/dev/null 2>&1
[[ ! -d "$HOME/.claude-mywork" ]]; assert_eq 0 $? "dir deleted"
[[ -z "$(find "$HOME" -maxdepth 1 -name '.claude-
mywork.removed.*' 2>/dev/null)" ]]; assert_eq 0 $? "no
archive sibling"

it "claude-remove-account rejects unknown alias"
( run_remove_account --alias nonexistent --yes >/dev/null 2>&1 )
rc=$?
[[ $rc -ne 0 ]]; assert_eq 0 $? "unknown alias errors out"

summary

```

8.15 tests/test_list.sh

```

#!/usr/bin/env bash
# test_list.sh - claude-list-accounts.sh reports accurate state.

set -uo pipefail

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"

source "$TESTS_DIR/lib/assert.sh"
source "$TESTS_DIR/lib/sandbox.sh"

make_sandbox

# Empty-host case first.
it "lists nothing when no accounts exist"
out="$(run_list_accounts 2>&1)"
[[ "$out" == *"ALIAS"* ]]; assert_eq 0 $? "header always printed"
echo "$out" | grep -E '^-' >/dev/null
[[ $? -ne 0 ]]; assert_eq 0 $? "no account rows"

acct1="$(make_account acct1)"
acct2="$(make_account acct2)"
run_unify >/dev/null 2>&1
run_add_account --alias claude3 --yes >/dev/null 2>&1

it "shows every account dir with creds + link status"

```

```

out="$(run_list_accounts 2>&1)"
assert_file_contains <(printf '%s\n' "$out") ".claude-acct1"
    "acct1 listed"
assert_file_contains <(printf '%s\n' "$out") ".claude-acct2"
    "acct2 listed"
assert_file_contains <(printf '%s\n' "$out") ".claude-claude3"
    "claude3 listed"
assert_file_contains <(printf '%s\n' "$out") "claude3" "claude3
    alias resolved"

it "reports the shared store path"
out="$(run_list_accounts 2>&1)"
assert_file_contains <(printf '%s\n' "$out") "$SHARED_DIR"
    "shared store path printed"

it "flags accounts with missing credentials"
# claude3 was created by add-account; it has no .credentials.json.
out="$(run_list_accounts 2>&1)"
line="$(echo "$out" | grep ".claude-claude3" || true)"
[[ "$line" == *" no *" ]]; assert_eq 0 $? "claude3 creds:no"
line1="$(echo "$out" | grep ".claude-acct1" || true)"
[[ "$line1" == *" yes *" ]]; assert_eq 0 $? "acct1 creds:yes"

summary

```

8.16 tests/test_export.sh

```

#!/usr/bin/env bash
# test_export.sh - claude-export-docs.sh produces HTML and PDF,
expands
# <!-- INCLUDE: path --> markers, and the output PDF starts with
"%PDF-".

set -uo pipefail

TESTS_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
SCRIPTS_DIR="$(cd "$TESTS_DIR/.." && pwd)"

source "$TESTS_DIR/lib/assert.sh"
source "$TESTS_DIR/lib/sandbox.sh"

make_sandbox
mkdir -p "$HOME/Documents/scripts"

# Tiny fixture markdown + a fixture script to include. We point
the
# export script at this synthetic doc instead of the real one.
fixture_script="$HOME/Documents/scripts/sample.sh"
cat > "$fixture_script" <<'EOF'
#!/usr/bin/env bash
echo "SAMPLE_SCRIPT_BODY_MARKER"
EOF

```



```

fixture_md="$HOME/Documents/fixture.md"
cat > "$fixture_md" <<'EOF'
---
title: Fixture
---
# Fixture

Body paragraph above the include.

```bash
<!-- INCLUDE: scripts/sample.sh -->

```

End of fixture. EOF

```

it "MD_FILE env override is honored and HTML is generated"
MD_FILE="$fixture_md" "$SCRIPTS_DIR/claude-export-docs.sh"
>/dev/null 2>&1 rc=$? assert_eq 0 "$rc" "export exits 0"
assert_file "$HOME/Documents/fixture.html" "html generated"
assert_file "$HOME/Documents/fixture.pdf" "pdf generated"

```

```

it "markers are expanded in output" assert_file_contains
"$HOME/Documents/fixture.html"
"SAMPLE_SCRIPT_BODY_MARKER" "include expanded"
assert_file_not_contains "$HOME/Documents/fixture.html" "<!--
INCLUDE:" "markers removed"

```

```

it "PDF starts with the %PDF- magic bytes" head -c 5 "$HOME/
Documents/fixture.pdf" | grep -q '^%PDF-' assert_eq 0 $? "PDF
magic header present"

```

```

it "regeneration is non-destructive (file timestamps move
forward)" MD_FILE="$fixture_md" "$SCRIPTS_DIR/claude-export-
docs.sh" >/dev/null 2>&1 assert_file "$HOME/Documents/
fixture.html" "html still present after rerun" assert_file "$HOME/
Documents/fixture.pdf" "pdf still present after rerun"

```

summary

---

## ## 9. Troubleshooting

Symptom	
Likely cause	
Fix	
-----	
-----	
-----	
`claude1` works, `claude2` shows empty memory	

```

Symlinks weren't created in `~/claude-milos85vasic2nd/`. | Re-
run `claude-unify.sh` – it's idempotent.
|
| New account can't see history |
Forgot to reload shell after `claude-add-account.sh`. |
`exec $SHELL -l` or `source ~/.local/share/claude-multi-account/
aliases.sh`. |
| Plugin loads fail with "plugin not found" |
Manifest path points outside the shared store. | Re-
run `claude-unify.sh` – the path-rewrite pass will fix it.
|
| Both accounts see *each other's* drafts in the queue |
`sessions/` is shared and two processes are running at once. |
Don't run two accounts concurrently, or remove `sessions` from
`SHARED_ITEMS`. |
| `pandoc` fails to find a PDF engine |
No weasyprint/wkhtmltopdf/chromium installed. |
`brew install weasyprint` or `apt install
weasyprint`. |
| Want to start completely over |
- |
`claude-rollback` then re-run
`install.sh`. |

```

---

\*Generated 2026-05-26. Maintain by editing this markdown file and re-running `claude-export-docs.sh`.\*